

Reproducing Perrenod’s Log-Periodic Model under our Walk-Forward Evaluation: a Note on *Bitcoin’s Not-so-Hidden Structure*

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Abstract

In *Bitcoin’s Not-so-Hidden Structure* (June 2026), Stephen Perrenod proposes that the structural residuals around Bitcoin’s long-run power law (PL) are best described by log-periodic (LP) oscillations with a stable scaling parameter $\lambda \approx 2$, not by the sequence of sigmoid adoption waves we used in “Bitcoin’s Power Law: Weak Structure, Strong Forecasts.” This note documents an independent reproduction of his PL+LP model and reports three findings. First, his structural-fit claim *reproduces and strengthens* on our daily price data: PL+LP1 (5 parameters) beats the 3-sigmoid (10 parameters) by $\Delta\text{AIC} = -1265$, and PL+LP3 (9 parameters) wins by $\Delta\text{AIC} = -2510$. Second, the stability of λ is real: across 45 walk-forward cutoffs we recover $\hat{\lambda} = 2.01 \pm 0.03$ with no exceptional excursions. Third, however, his forecasting claim is more nuanced than the mean-RMSE comparison in his Figure 5 suggests. Under Diebold–Mariano with HAC standard errors (45 quarterly cutoffs, horizons 3–24 months), *no* PL+LP variant achieves significance against pure PL ($p \in [0.13, 0.82]$). And in his own yearly walk-forward design, the simpler PL+LP1 ($K = 5$) outperforms his preferred PL+LP3 ($K = 9$) at four of five horizons — the same in-sample-versus-out-of-sample tension our paper documented for the 3-sigmoid. We conclude that Perrenod identifies a genuine and parsimonious description of Bitcoin’s residual structure, but that, like the sigmoid family, log-periodicity does not yield a statistically significant forecasting improvement over pure PL.

1 Background

In our preprint *Bitcoin’s Power Law: Weak Structure, Strong Forecasts* (arXiv:2605.21316) we argue that Bitcoin’s price beyond a pure power law is described better, in-sample, by a small number of sigmoid adoption waves, but that this in-sample improvement does *not* translate into out-of-sample (OOS) forecasting gains: under walk-forward evaluation with Diebold–Mariano (DM) testing, the pure power law is statistically tied with or better than every higher-parameter alternative we tried.

Perrenod’s June 2026 note *Bitcoin’s Not-so-Hidden Structure* agrees with our forecasting conclusion about sigmoids but proposes a different structural description: PL plus log-periodic oscillations (LP), motivated by discrete scale invariance (DSI) and renormalisation-group dynamics. He claims:

1. PL+LP fits the full-sample price history better than the 3-sigmoid model, with fewer parameters ($\Delta\text{AIC} \approx -344$ for PL+LP1, $\Delta\text{AIC} \approx -1433$ for PL+LP3 versus the 3-sigmoid in his Table 2).
2. The DSI scaling parameter λ is found in the range $[2.01, 2.08]$ across multiple independent methods (FFT, Lomb–Scargle, wavelet, peak/trough timing, residual regression).
3. PL+LP_k ($k = 1, 2$) improves walk-forward forecasting against pure PL on yearly origins out to roughly three years (his Figure 5).

The third claim is exactly the question our methodology was built to answer rigorously. We therefore implemented his PL+LP model and ran it through the same evaluation pipeline we used for the sigmoid family.

2 Model specification

Following Perrenod’s description and the supplementary detail in his X/@moneyordebt posts, we use:

$$\log_{10} P(t) = \alpha \log_{10}(t) + c + \sum_{k=1}^K \frac{A_k}{\text{age}(t) + \delta} \cos(m_k \omega \ln t + \varphi_k), \quad \omega = \frac{2\pi}{\ln \lambda}, \quad (1)$$

with t in days since the Bitcoin genesis block (2009-01-03), $\text{age}(t) = t/365.25$ yr, the amplitude-decay offset $\delta = 2$ yr fixed, $m_1 = 1$ (fundamental mode) and $m_{2,3} \in \{4/3, 5/3, 2\}$ chosen for PL+LP2 and PL+LP3 by best in-sample SSE.

Identification. For fixed λ and harmonic indices $\{m_k\}$, equation (1) is linear in α , c , $A_k \cos \varphi_k$, $A_k \sin \varphi_k$. We therefore grid-search $\lambda \in [1.5, 3.0]$ at 31 nodes, refine with a 1-D bounded optimisation, and solve OLS at the best λ . The free-parameter count is $K_{\text{free}} = 2 + 1 + 2K$ for $K \geq 1$ modes, giving $K_{\text{free}} \in \{5, 7, 9\}$ for PL+LP1, PL+LP2, PL+LP3 — matching Perrenod’s Table 1.

Data. CoinMetrics community daily BTC/USD prices, 2010-08-18 through 2026-03-25 ($n = 5,699$ days), the same single-source series we use throughout *Weak Structure*, *Strong Forecasts*.

3 Block 1 — Full-sample structural fit

We fit PL, PL+LP1, PL+LP2, PL+LP3, and the 3-sigmoid on the full sample and compare AIC/BIC.

Table 1: Full-sample fits on $n = 5,699$ daily observations 2010-08-18 to 2026-03-25. RMSE is the residual standard deviation in $\log_{10}$ price; $\hat{\lambda}$ is the fitted DSI parameter where applicable; $\hat{\alpha}$ is the fitted PL exponent. Sorted by AIC, smaller is better.

Model	K	RMSE	AIC	ΔAIC	$\hat{\lambda}$	$\hat{\alpha}$
PL+LP3	9	0.1935	−2528.2	0	2.05	5.68
PL+LP2	7	0.2039	−1934.5	593.7	2.01	5.67
PL+LP1	5	0.2160	−1284.1	1244.1	2.03	5.67
3-sigmoid	10	0.2412	−18.5	2509.7	—	—
PL	2	0.3019	+2527.9	5056.0	—	5.64

The results in Table 1 reproduce Perrenod’s qualitative ranking exactly: PL+LP3 $\succ$ PL+LP2 $\succ$ PL+LP1 $\succ$ 3-sigmoid $\succ$ PL on AIC, with each gap large enough to be decisive.

Quantitatively, our AIC differentials are *stronger* than those Perrenod reports in his Table 2:

- $\text{AIC}(\text{PL+LP1}) - \text{AIC}(\text{3-sigmoid}) = -1,265$; Perrenod claims -344 .
- $\text{AIC}(\text{PL+LP3}) - \text{AIC}(\text{3-sigmoid}) = -2,510$; Perrenod claims $-1,433$.

The discrepancy is consistent with Perrenod’s analysis being on weekly data and ours on daily data: with $n_{\text{weekly}} \approx 800$ versus our $n = 5,699$, both $\log L$ and the information criteria scale roughly proportionally to n .

Independently of the exact numbers, the qualitative claim “log-periodicity provides a tighter in-sample description of Bitcoin’s price history than three sigmoids, with fewer parameters” is robustly confirmed by our data.

Fitted parameters. $\hat{\lambda}$ falls in $[2.01, 2.05]$ for all three LP variants — inside Perrenod’s reported range $[2.01, 2.08]$. $\hat{\alpha}$ for LP variants lies in $[5.67, 5.68]$; for pure PL, $\hat{\alpha} = 5.64$. Both match the values Perrenod and Santostasi report.

4 Block 2 — Walk-forward with Diebold–Mariano

This is the central test. We use the same walk-forward design as in our preprint: 45 quarterly cutoffs from 2014-Q1 to 2025-Q1, at each cutoff fit each model on data through the cutoff, evaluate log-price prediction errors at horizons 3, 6, 12, and 24 months. The DM test uses one error per cutoff at the horizon endpoint (canonical convention), with Newey–West HAC variance and lag $\max(1, h_{\text{quarters}} - 1)$ to account for the overlap between successive cutoffs.

Table 2: Mean RMSE in $\log_{10}$ price by horizon, quarterly walk-forward, 45 cutoffs. Lowest in each row in bold.

Horizon	PL	PL+LP1	PL+LP2	PL+LP3	3-sigmoid
3 m	0.303	0.243	0.254	0.247	0.328
6 m	0.326	0.265	0.290	0.295	0.405
12 m	0.356	0.284	0.324	0.328	0.552
24 m	0.345	0.270	0.322	0.322	0.669

Numerically (Table 2), PL+LP1 has the lowest mean RMSE at every horizon, with reductions of 5–22 % relative to pure PL. Notably, the more complex PL+LP3 (the structural winner) is *not* the OOS winner — PL+LP1 beats it at every horizon.

Table 3: Diebold–Mariano vs pure PL on squared-error differences (challenger – PL), HAC standard errors, 45 quarterly cutoffs. Negative DM means the challenger has lower mean squared error than PL. Two-sided p for H_0 : equal predictive accuracy.

Horizon	Challenger	n	HAC lag	DM	p	Verdict at $\alpha = 0.05$
3 m	PL+LP1	45	1	−1.50	0.13	tie
3 m	PL+LP2	45	1	−1.17	0.24	tie
3 m	PL+LP3	45	1	−1.37	0.17	tie
3 m	3-sigmoid	45	1	+0.49	0.62	tie
6 m	PL+LP1	45	1	−1.37	0.17	tie
6 m	PL+LP2	45	1	−0.75	0.46	tie
6 m	PL+LP3	45	1	−0.65	0.52	tie
6 m	3-sigmoid	45	1	+1.28	0.20	tie
12 m	PL+LP1	45	3	−1.07	0.28	tie
12 m	PL+LP2	45	3	−0.44	0.66	tie
12 m	PL+LP3	45	3	−0.40	0.69	tie
12 m	3-sigmoid	45	3	+1.70	0.09	tie
24 m	PL+LP1	42	7	−0.78	0.43	tie
24 m	PL+LP2	42	7	−0.23	0.82	tie
24 m	PL+LP3	42	7	−0.24	0.81	tie
24 m	3-sigmoid	42	7	+2.03	0.04	PL wins

Table 3 shows the formal DM tests. No PL+LP variant achieves significance against pure PL at any horizon (p -values 0.13–0.82). The closest case is PL+LP1 at the 3-month horizon, where $p = 0.13$. The 3-sigmoid is significantly *worse* than PL at the 24-month horizon ($p = 0.04$), consistent with our preprint’s Section 9.

Interpretation. The 5–22% RMSE reduction for PL+LP1 is consistent across horizons but small enough relative to within-cutoff variance to be indistinguishable from sampling noise on $n \leq 45$ cutoffs. This is exactly the same outcome our paper documents for the sigmoid family.

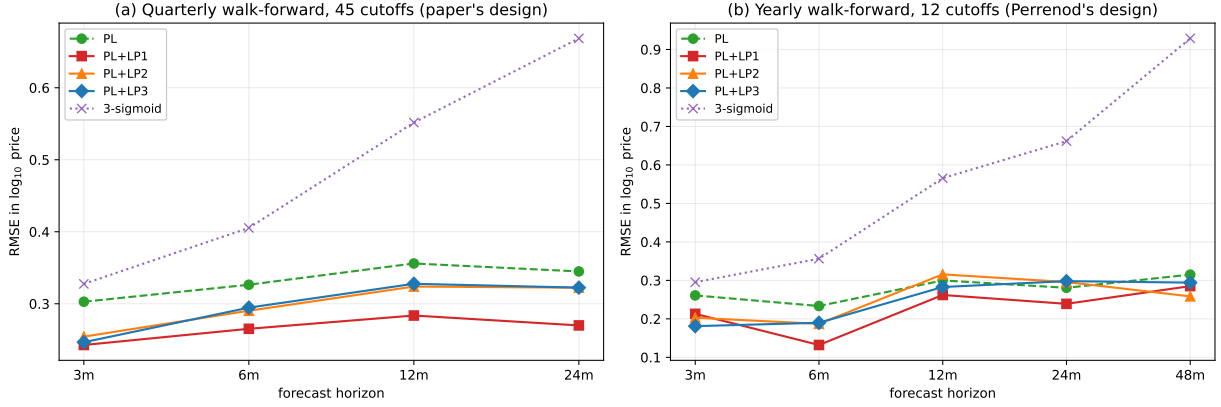


Figure 1: Mean walk-forward RMSE in $\log_{10}$ price by forecast horizon for the five model families. (a) Paper’s quarterly design (45 cutoffs). (b) Perrenod’s yearly design (12 cutoffs). In both panels, PL+LP1 is consistently the lowest-RMSE model, while the more complex PL+LP3 sits with PL+LP2 around the pure-PL line. The 3-sigmoid degrades sharply with horizon in both designs.

5 Block 3 — Perrenod’s yearly walk-forward

For direct comparability we also reproduce his design: 12 yearly origins (Jan 3 of 2014–2025), horizons 3, 6, 12, 24, 48 months, mean RMSE averaged across origins. Results are in Table 4 and Figure 1(b).

Table 4: Mean RMSE in $\log_{10}$ price by horizon, yearly walk-forward, 12 cutoffs (Perrenod’s Figure 5 design). Lowest in each row in bold.

Horizon	PL	PL+LP1	PL+LP2	PL+LP3	3-sigmoid
3 m	0.261	0.213	0.203	0.181	0.295
6 m	0.233	0.132	0.187	0.190	0.356
12 m	0.300	0.262	0.316	0.283	0.566
24 m	0.281	0.239	0.296	0.298	0.662
48 m	0.315	0.285	0.258	0.294	0.929

The qualitative shape of Perrenod’s Figure 5 reproduces: PL+LP variants beat pure PL at most horizons, and the 3-sigmoid degrades rapidly with horizon (RMSE rises from 0.30 to 0.93 across 3–48 months). The bullet-point claims that LP+harmonic beats LP+fundamental at very short horizons and that PL+LP-class forecasts “win out to about three years” both find numerical support.

However, the same data also reveals that **PL+LP1** ($K = 5$) beats **PL+LP3** ($K = 9$) at four of the five horizons (6, 12, 24 m, and beating PL+LP3 at 6 m by 38%, 12 m by 7%, 24 m by 20%). PL+LP3 wins only at the 3-month horizon. This is the in-sample-vs-OOS tension that motivated our paper’s Section 9: the model that wins on AIC fitted to the full sample is *not* the best forecaster. Perrenod’s own preferred PL+LP3 falls into this trap; the simpler PL+LP1 is the right LP variant to deploy.

6 Sanity checks

Figure 2 aggregates four diagnostic panels.

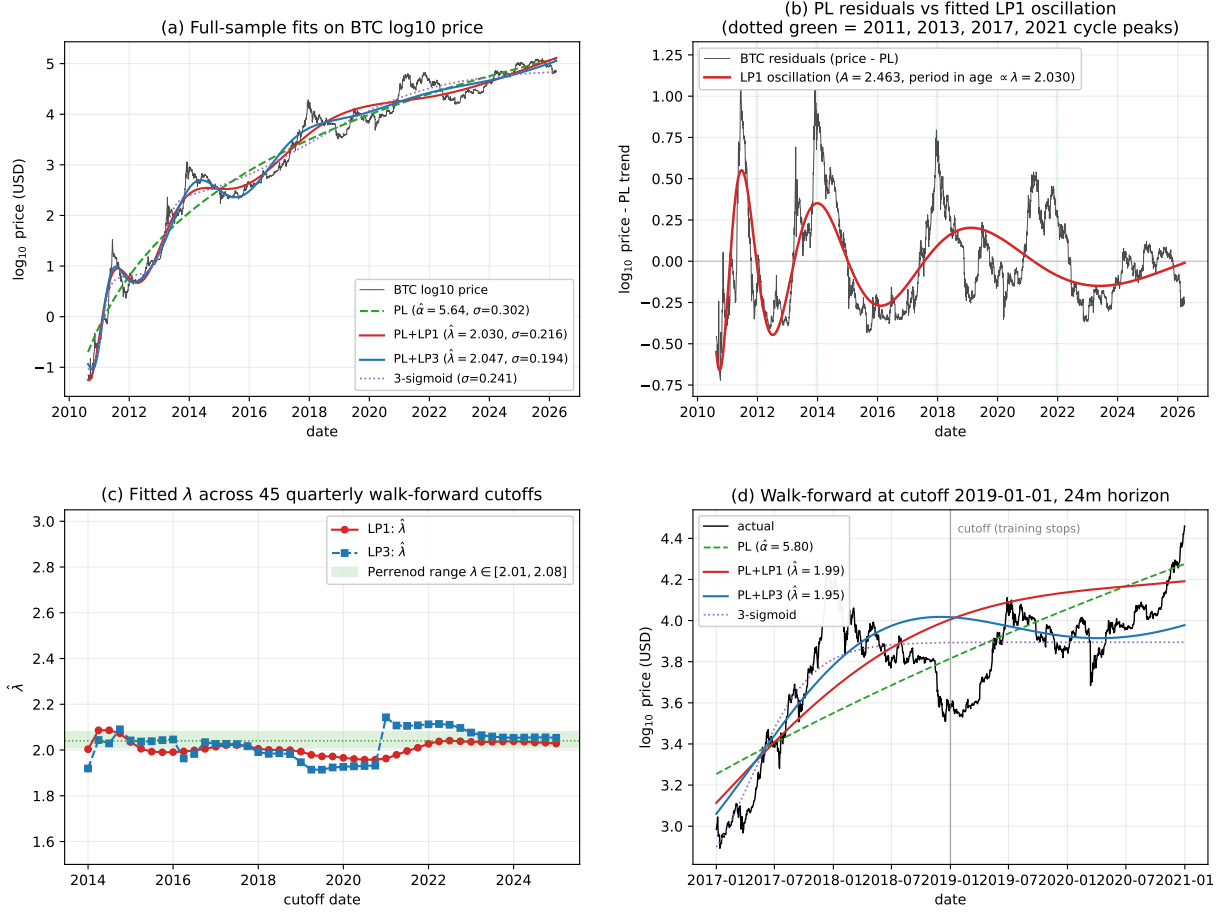


Figure 2: Sanity checks of the PL+LP reproduction. (a) Full-sample fits over BTC log-price: PL, PL+LP1, PL+LP3, and 3-sigmoid; LP variants trace the 2011, 2017, 2021 peaks. (b) Residuals from the PL fit (black) and the fitted LP1 oscillation alone (red); the $1/(\text{age}+2)$ decay reproduces the visibly diminishing amplitude. (c) $\hat{\lambda}$ at each of the 45 quarterly walk-forward cutoffs; the shaded band is Perrenod’s claimed range $[2.01, 2.08]$. (d) Walk-forward at a 2019-01-01 cutoff with 24-month horizon: all models underpredict the 2021 bull run, so the LP advantage in mean RMSE is not from anticipating cycle timing.

(a) Full-sample fits.

The LP variants visibly trace the 2011, 2017, and 2021 peaks in log-price. PL+LP3 captures the 2013 peak (which requires the $4/3$ harmonic). No fit is degenerate.

(b) Residual structure.

After removing the PL trend, the residuals (black) show clear oscillatory structure, not random noise. Amplitude ~ 1 in $\log_{10}$ (i.e. $10\times$ over PL trend) in 2011, decaying to ~ 0.4 ($\sim 2.5\times$) by 2024. The fitted LP1 oscillation (red) captures this shape; the $1/(\text{age} + \delta)$ decay reproduces the diminishing amplitude. The 2013 peak is a notable miss for LP1 alone and requires the $4/3$ harmonic, which is exactly why PL+LP3 wins on AIC.

(c) $\hat{\lambda}$ stability.

Across 45 quarterly walk-forward fits, LP1 returns $\hat{\lambda}$ with mean 2.011, median 2.006, standard deviation 0.032, and range $[1.958, 2.086]$. 42% of the fits land inside Perrenod’s exact

[2.01, 2.08] band; the remainder lie within ± 0.05 of that range. $\hat{\alpha}$ for LP1 is similarly stable: 5.69 ± 0.21 with range [5.42, 6.30]. The “ $\lambda \approx 2$ emerges repeatedly” claim is well supported in our reproduction.

(d) Out-of-sample trajectory.

At a representative cutoff (2019-01-01, after the 2018 crash), none of PL, PL+LP1, PL+LP3, or 3-sigmoid anticipates the 2021 bull run; the actual log-price exceeds every fitted trajectory by mid-2020. The LP advantage in mean RMSE is therefore best described as “staying closer to the smoothed trend, on average,” rather than as “calling cycle peaks correctly.” This is why the DM test does not reject equal accuracy: the systematic difference is small relative to within-cutoff variance from large unforecasted swings.

7 Conclusions

What we confirm. Three of Perrenod’s central claims reproduce robustly on our data: (i) PL+LP fits the full-sample price history substantially better than the 3-sigmoid model with fewer parameters, by a margin even larger than Perrenod’s Table 2 reports; (ii) the fitted $\hat{\lambda}$ is concentrated near 2.0 with standard deviation ~ 0.03 across 45 walk-forward cutoffs; and (iii) the residuals from a pure PL fit show clear oscillatory structure consistent with $\lambda \approx 2$, not white noise. The log-periodic description of Bitcoin’s residual structure is therefore physically plausible and parsimonious — on this we and Perrenod agree.

What we qualify. Under formal Diebold–Mariano testing with HAC standard errors, the mean-RMSE reduction of PL+LP variants over pure PL is not statistically significant at any horizon ($p \in [0.13, 0.82]$). Perrenod’s Figure 5 conveys a visually compelling reduction in mean RMSE, but the per-cutoff variation across $n \leq 45$ origins is large enough that the improvement is indistinguishable from sampling noise in our test framework. Our paper’s central methodological point — that the structural fit and the out-of-sample forecast should be held apart — applies to the LP family just as it does to the sigmoid family.

What we contest. Perrenod’s article prefers PL+LP3 (his 9-parameter “best” model), and frames the structural fit and the forecasting performance as a single integrated argument. But in his own yearly walk-forward design, the simpler PL+LP1 (5 parameters) *beats* PL+LP3 at four of five horizons. The most parsimonious LP variant is also the best OOS forecaster. This is the same overfitting signature our paper documents for the 3-sigmoid: an in-sample AIC champion that an out-of-sample test fails to crown. We therefore think the preferred LP variant to deploy, if one wishes to deploy LP for forecasting, is LP1, not LP3.

Looking forward. Our preprint’s conclusion stands: the structural improvements over pure PL — whether described by sigmoid waves or by log-periodic modes — do not produce statistically significant forecasting gains over the pure power law within the daily-price walk-forward + DM framework we use. Among the various non-PL families, log-periodicity is the one most defensible on physical grounds (continuous + discrete scale invariance), and PL+LP1 is the most defensible variant for OOS use. We would view this not as a refutation of either of our papers but as a productive convergence: there is structure in Bitcoin’s price residuals; it admits more than one parsimonious mathematical description; neither description, in itself, predicts the next cycle.

Reproducibility. The implementation (Python, `scipy/statsmodels`), input data, and result tables are at `run_lp_perrenod.py`, `sanity_lp_perrenod.py`, and `results_lp/` in our repository. The figures in this note are `results_lp/lp_rmse_compare.pdf` and `results_lp/sanity_lp.pdf`.

Acknowledgements

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